

Special Problem Proposal: Early Detection of the Risk of
Diabetes Using Calibrated Explanations and Explainable
AI (XAI) with Philippine Data

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1 Introduction

1.1 Background of the Study

Diabetes is a chronic disease characterized by elevated blood sugar levels and remains a significant public health concern in the Philippines. Individuals with diabetes face higher risks of cardiovascular disease, kidney failure, nerve damage, and vision impairment [1]. Recent national data reveal that the prevalence of diabetes among Filipino adults has reached approximately 7.5% as of 2024, with over 4.7 million cases reported and an alarming number of undiagnosed individuals, surpassing 53% of the diabetes-affected population [2]. In 2020 alone, diabetes accounted for 37,265 deaths, ranking as the fourth leading cause of mortality in the country [3]. The increasing urbanization, sedentary lifestyles worsened by COVID-19 lockdowns, and changing dietary patterns have accelerated obesity and other risk factors among Filipinos, intensifying the diabetes epidemic [4].

Despite numerous international studies on diabetes risk assessment, most rely on data from Western or global populations [5], failing to account for the unique Filipino demographic, genetic, and environmental factors [4]. This reliance on international datasets undermines the applicability and reliability of predictive models for local populations, which presents a distinctive research gap. The need for a customized, locally relevant platform for accurate diabetes risk prediction is critical to improving public health outcomes [1]. Recent advancements in artificial intelligence (AI) and machine learning have opened new avenues for medical diagnostics and predictive healthcare. However, transparency and interpretability remain major challenges for clinical adoption. Explainable Artificial Intelligence (XAI) bridges this gap by providing interpretable predictions that empower healthcare practitioners and patients alike.

In the Philippine context, there is a pressing need for predictive models tailored to local data to ensure confidence in clinical decisions. This study addresses this gap by developing a machine learning system that integrates three key components: comprehensive evaluation of multiple machine learning models (including tree-based algorithms and deep learning approaches) trained exclusively on Philippine-specific data to identify the optimal predictor; Explainable Artificial Intelligence (XAI) techniques, specifically LIME and SHAP, to provide granular interpretations of predictions; and post-hoc cali-

bration mechanisms to align predicted probabilities with actual clinical risk levels. By systematically combining rigorous model comparison, interpretability, and probabilistic calibration, this research aims to foster robust clinical confidence, thereby reducing undiagnosed cases and supporting effective diabetes prevention strategies within Filipino communities.

1.2 Statement of the Problem

Diabetes is a significant health challenge in the Philippines, and machine learning models have shown promise in predicting its risk. While Explainable Artificial Intelligence (XAI) techniques, such as LIME and SHAP, help provide transparency into model predictions, they often fail to address the issue of model calibration. Uncalibrated models produce probability estimates that may be overconfident or underconfident, leading to unreliable risk predictions. This undermines the potential of AI to support clinical decision-making and reduce diabetes incidence. Therefore, there is a critical need for calibration in XAI to ensure that predicted probabilities align with actual risks. By integrating calibration techniques into XAI models, this study aims to improve both the trustworthiness and interpretability of diabetes risk predictions. This research will contribute to developing a Calibrated Explainable AI model tailored to Philippine data, ultimately enhancing predictive accuracy and fostering greater confidence in AI-driven healthcare applications.

1.3 Research Questions

This study aims to answer the following research questions:

1. Which among the evaluated machine learning models — Naive Bayes, k-Nearest Neighbors, Random Forest, XGBoost, LightGBM, CatBoost, AdaBoost, Bayesian-Optimized TabNet, and ensemble methods (Voting, Stacking) — achieves the best diabetes risk prediction performance on the DOST-FNRI 2018–2021 Expanded National Nutrition Survey dataset, as measured by F2-score and AUPRC?
2. How do probability calibration techniques (Platt Scaling and Isotonic Regression) affect the reliability of predicted diabetes risk probabilities from the best-performing model, as assessed by Brier Score, Expected Calibration Error (ECE), and calibration plots?

3. What are the most influential clinical, anthropometric, biochemical, and dietary features driving individual diabetes risk predictions in the Philippine context, as revealed by SHAP and LIME explanations applied to the best-performing calibrated model?
4. How can the best-performing calibrated model with XAI explanations be effectively delivered through a web-based decision support tool that provides real-time diabetes risk assessment with interpretable, probability-calibrated outputs for end-users?

1.4 Objectives of the Study

The study aims to facilitate early detection and prediction of diabetes risk among individuals in the Philippines. Calibration techniques will be applied to improve the reliability of predicted probabilities, and XAI methods will be integrated to provide interpretable explanations. The specific objectives of this study are as follows:

1. Allows the primary user (healthcare practitioner or end-user) to:
 - (a) Input personal health information relevant to diabetes risk prediction (e.g., age, gender, medical history, anthropometric and biochemical measurements).
 - (b) Request a prediction of the likelihood of developing diabetes based on the input data.
 - (c) Receive an interpretable explanation of the prediction through XAI methods, specifically SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), to help understand the key factors influencing the model's output.
 - (d) View the model's output in the form of a calibrated diabetes risk probability and risk level classification, supporting informed decision-making and preventive actions.
2. Allows the system to:
 - (a) Perform comprehensive data pre-processing on the DOST-FNRI 2018–2021 ENNS dataset, including missing value imputation, target variable encoding,

feature scaling (MinMaxScaler), correlation-based feature elimination, and class imbalance handling (SMOTE).

- (b) Apply hyperparameter tuning via RandomizedSearchCV (optimizing for F2-score) to the following models: k-Nearest Neighbors, AdaBoost, XGBoost, Random Forest, LightGBM, and CatBoost; and Bayesian Optimization via Optuna for TabNet.
- (c) Evaluate and compare the performance of all trained models using Precision, Recall, F2-score, AUPRC, and ROC-AUC, where F2-score serves as the primary metric.
- (d) Implement and compare the following machine learning models:
 - i. Traditional ML: Naive Bayes, k-Nearest Neighbors, Random Forest, XGBoost, LightGBM, CatBoost, AdaBoost
 - ii. Deep Learning: Bayesian-Optimized TabNet (BO-TabNet)
 - iii. Ensemble Methods: Soft Voting Ensemble, Stacking Ensemble (with Logistic Regression meta-learner)
- (e) Apply both Platt Scaling and Isotonic Regression to calibrate the predicted probabilities of trained models using a held-out validation set, and evaluate calibration quality using Brier Score and Expected Calibration Error (ECE).
- (f) Apply SHAP and LIME to the best-performing calibrated model to generate global feature importance rankings and local per-instance explanations.

3. Allows the user to:

- (a) Input a manual entry or upload a CSV file containing relevant health information for diabetes risk prediction.
- (b) Request a diabetes risk prediction with calibrated probability estimates and XAI explanations.
- (c) View the prediction results including:
 - i. Calibrated risk probability (e.g., “65% diabetes risk”)
 - ii. Risk level classification (Low, Moderate, or High)
 - iii. SHAP or LIME feature importance explanations indicating which factors most influenced the prediction

1.5 Significance of the Project

This study aims to provide a **reliable and calibrated explainable AI (XAI) model** for accurate and early prediction of diabetes risk, which will significantly contribute to the following:

1. Prediabetic Patients

This study's *calibrated XAI model* can provide actionable insights for prediabetic individuals, helping them make informed decisions regarding lifestyle changes to prevent or manage diabetes. The reliable predictions and interpretations provided by the model will empower patients to take timely actions toward healthier lifestyles and more proactive health management.

2. Medical Professionals

By integrating calibration with XAI, the study provides *accurate, interpretable, and trustworthy* diabetes risk predictions that assist healthcare professionals in making informed clinical decisions. The ability to trust the *probability estimates* generated by the model will facilitate personalized care, early intervention, and better patient outcomes. The explainability of the model also ensures that clinicians understand the reasoning behind the predictions, enhancing trust in the AI system.

3. Future Researchers in Health and Machine Learning

This study contributes to the field of healthcare AI by integrating calibration techniques into XAI models, making it a key reference for future researchers. The findings will offer insights into how calibration improves the reliability and clinical applicability of AI models, guiding future work in healthcare AI, particularly in ensuring that AI-based predictions are not only accurate but also trustworthy and interpretable.

1.6 Scope and Limitations

The following are the limitations of this study:

1. This study will only utilize local datasets of **Department of Science and Technology - Food and Nutrition Research Institute (DOST-FNRI)** to develop

and validate a model for predicting the risk of diabetes.

2. The study will focus on developing and validating a *Calibrated Explainable AI (XAI)* model using machine learning techniques such as tree-based algorithms and deep learning approaches, integrated with calibration methods like Platt scaling and Isotonic Regression, to ensure accurate and reliable risk predictions.
3. Limitations in the quality and quantity of available Philippine health datasets may impact the model’s performance, *especially when trying to represent diverse regions or populations*.
4. While XAI techniques like LIME will be used to enhance the *interpretability* of the model, some features of the model may still be difficult to explain comprehensively, potentially influencing how users (healthcare professionals and researchers) interpret model predictions.
5. The *calibrated XAI* model developed in this study is intended as a *decision support tool* to assist healthcare professionals with diabetes risk prediction. It is not intended to replace or override professional medical judgment or clinical decisions.
6. The trained model’s predictive capacity is limited to *predicting the general risk of diabetes* and does not have the ability to distinguish between *type 1 diabetes* and *type 2 diabetes* in individual predictions.
7. Calibration techniques applied on a dataset from the **Philippine population** may not generalize well to other populations or settings. This is because the calibration process is specific to the data used to train the model, and demographic, lifestyle, or healthcare-related differences in other regions could lead to miscalibrated predictions. Further research and cross-validation across diverse populations may be necessary.

1.7 Assumptions

1. It is assumed that the DOST-FNRI 2018–2021 Expanded National Nutrition Survey (ENNS) dataset contains sufficient records and feature coverage across clinical, anthropometric, biochemical, and dietary components to train machine learning models capable of generalizing to the adult Filipino population.

2. It is assumed that the selected machine learning algorithms are capable of learning meaningful patterns from the preprocessed ENNS dataset for binary diabetes risk classification, and that hyperparameter tuning via RandomizedSearchCV and Bayesian Optimization will yield models that perform better than default-parameter baselines.
3. It is assumed that applying Platt Scaling or Isotonic Regression to a held-out validation set will produce calibrated probability estimates that are more reliable than the raw uncalibrated outputs, as indicated by improvements in Brier Score and Expected Calibration Error (ECE), even if perfect calibration is not achievable.
4. It is assumed that SHAP and LIME will produce sufficiently interpretable local and global explanations to identify the primary features influencing individual diabetes risk predictions. It is acknowledged that LIME explanations may exhibit instability across similar instances and that neither method guarantees clinically definitive explanations.
5. It is assumed that the calibrated probability outputs of the best-performing model can serve as meaningful decision-support information for end-users, while recognizing that clinical validation by qualified medical professionals remains necessary before any deployment in actual healthcare settings.
6. It is assumed that the best-performing trained model, once serialized, can be loaded and served within a Python-based web application framework to provide predictions within a reasonable response time under standard hardware conditions.

2 Review of Related Literature

This chapter presents a comprehensive review of existing literature relevant to diabetes prediction using machine learning and explainable artificial intelligence, with particular emphasis on Philippine-specific contexts and datasets. The review is organized into seven major themes: the burden of diabetes in the Philippines, available Philippine datasets for diabetes prediction, machine learning models for diabetes classification, explainable AI methods, calibrated explanations, probability calibration techniques, and evaluation metrics for imbalanced datasets.

2.1 Burden of Diabetes in the Philippines

Diabetes mellitus represents a critical and escalating public health challenge in the Philippines, with profound implications for mortality, morbidity, and healthcare costs. The burden of diabetes has reached alarming proportions, with an estimated 4.3 million Filipinos diagnosed with diabetes and an additional 2.8 million remaining undiagnosed in 2021. The age-standardized prevalence stands at 7.5% among adults aged 20-79 years as of 2024, with projections indicating an increase to 8.6 million cases by 2050. This represents a substantial increase from 1.2 million cases in 2000, underscoring the rapidly growing epidemic.

The mortality burden is equally concerning. Diabetes mellitus consistently ranks among the top five leading causes of death in the Philippines. In 2024 alone, diabetes was responsible for 43,944 deaths, accounting for approximately 6.3% of all deaths nationally. By March 31, 2025, diabetes-related mortality had already reached 31,321 cases, or 6.6% of total deaths. These figures highlight the urgent need for early detection and intervention strategies to prevent premature mortality.

The complications of diabetes further compound the disease burden in the Philippines. Diabetic retinopathy is identified as a leading cause of preventable blindness, particularly in Region 3 of the Philippines. Moreover, diabetic nephropathy contributes to 38% of all renal disease cases in the country, emphasizing the systemic nature of diabetes complications and their strain on healthcare resources.

The local burden is exacerbated by structural and socioeconomic challenges. Financial constraints, including limited insurance coverage and high out-of-pocket costs,

significantly impede access to diabetes care, amplifying the disease’s impact on vulnerable populations. The Filipino diet, typically high in refined carbohydrates, added sugar, and unhealthy fats, contributes to obesity, insulin resistance, and elevated blood glucose levels. Cultural practices centered around food celebrations further increase the risk of overconsumption of high-calorie and high-sugar foods.

A study analyzing the 8th Philippine National Nutrition Survey found that young Filipinos with diabetes exhibited higher risk profiles, including obesity class II prevalence of 22% compared to 12% in older adults, along with elevated rates of current drinking (56% vs 37%) and smoking (28% vs 18%). Additionally, 80% of both young-onset and late-onset diabetes patients presented with metabolic syndrome.

The COVID-19 pandemic exacerbated diabetes management challenges in the Philippines. Access to essential diabetes care, including routine monitoring and medication distribution, was significantly disrupted during quarantine periods. The temporary suspension of offloading devices and wound monitoring led to progression of diabetic foot ulcers to severe soft tissue infections requiring amputations. The intersection of diabetes care with pandemic-related disruptions underscores the fragility of healthcare systems and the necessity for resilient, accessible early detection mechanisms to prevent complications.

This substantial burden—characterized by high prevalence, significant mortality, debilitating complications, and socioeconomic barriers—establishes the critical need for early detection tools and interpretable prediction models that can be deployed effectively across diverse healthcare settings in the Philippines.

2.2 Philippine Datasets for Diabetes Risk Prediction

The availability and characteristics of Philippine datasets for diabetes risk prediction are crucial for developing locally relevant machine learning models. However, the literature reveals significant challenges and gaps in this domain.

The Department of Science and Technology-Food and Nutrition Research Institute (DOST-FNRI) conducts the Expanded National Nutrition Survey (ENNS), which provides comprehensive data on the eating habits, nutrition, and health status of Filipinos. The 2018-2020 ENNS was designed as a rolling survey covering 45,957 households in 2018 and 49,042 households in 2019 across various provinces and highly urbanized cities. This

dataset includes information on diabetes prevalence and related metabolic indicators. According to the 2021 ENNS, Filipino adults had a diabetes prevalence of 7.1% in the 20-79 age group, and the survey documented pre-diabetes status among the population. While DOST-FNRI datasets are valuable for nutritional and epidemiological research, access requires formal requests through the Freedom of Information portal.

Studies have utilized Philippine-specific datasets for diabetes prediction model development. One study optimized a bagged tree ensemble classifier using data from the 2019 Philippine National Nutrition Survey to predict diabetes prevalence in Filipino women. The model focused on six key predictors: pregnancies, glucose levels, blood pressure, BMI, age, and diabetes pedigree function.

Recent research developed an automated machine learning (AutoML) model for referable diabetic retinopathy using 2,000 ultrawide field (UWF) retinal images from local Philippine datasets. The dataset was labeled according to the Early Treatment Diabetic Retinopathy Study (ETDRS) severity grading and split with 80% for training, 10% for validation, and 10% for testing. The model achieved a sensitivity of 88% (95% CI: 0.80-0.94) and specificity of 83% (95% CI: 0.74-0.89) on external validation, approaching US FDA requirements for referable DR diagnosis.

A 2025 study employed a multi-method classification approach involving binomial logistic regression, K-means clustering, and decision tree analysis on 947 Filipino participants aged 24-79 years. The study identified BMI as the most significant predictor ($\chi^2(1) = 104.44$, $p < .001$), followed by HbA1c ($\chi^2(1) = 51.80$, $p < .001$), triglycerides ($\chi^2(1) = 12.44$, $p < .001$), and LDL ($\chi^2(1) = 9.15$, $p = .002$). The model demonstrated excellent predictive performance with a McFadden R^2 of 0.80 and Nagelkerke R^2 of 0.85. K-means clustering successfully categorized participants into two distinct groups based on biomarker profiles, differentiating those at higher and lower diabetes risk.

Despite these efforts, significant preprocessing challenges exist when working with Philippine datasets. Common issues include missing data, class imbalance (with diabetes cases typically representing a minority class), heterogeneous data sources, and the need for culturally appropriate feature engineering.

2.3 Machine Learning Models for Diabetes Prediction

The application of machine learning to diabetes prediction has evolved significantly, with various algorithms demonstrating varying degrees of success. Recent literature emphasizes the importance of model selection, hyperparameter optimization, and ensemble methods to achieve robust predictive performance.

2.3.1 Random Forest

Random Forest has demonstrated strong performance across multiple diabetes prediction studies. A 2024 study reported that Random Forest achieved an F1-score of 0.87 and accuracy of 89% on diabetes prediction tasks. The algorithm's effectiveness stems from its ability to handle both linear and non-linear relationships, reduce overfitting through ensemble averaging, and provide feature importance rankings that aid clinical interpretation. Remarkably, using only four features (general health, high blood pressure, BMI, and age), the model still achieved a robust F1-score of 0.751, suggesting that simpler models using fewer but high-impact features can be effectively deployed without sacrificing performance. A 2025 study comparing various algorithms found that Random Forest achieved 76.30% accuracy, ranking second after Neural Networks (78.57%) among evaluated algorithms.

2.3.2 K-Nearest Neighbors (KNN)

K-Nearest Neighbors is a non-parametric algorithm that classifies instances based on the majority class of their nearest neighbors in the feature space. A 2022 study addressing class imbalance in diabetes prediction using SMOTE-ENN resampling found that KNN achieved 98.38% accuracy, outperforming other methods including logistic regression (75.32%), decision tree (74.03%), random forest (72.73%), SVM (76.62%), gradient boosting (74.68%), and Naive Bayes (74.03%). This exceptional performance was achieved on the processed BRFSS dataset after applying the synthetic minority over-sampling technique integrated with the edited nearest neighbor method.

A comprehensive 2024 study investigating the optimal k and distance measures found that the best performance was achieved with $k=35$ and Euclidean distance, producing 80.5% accuracy on the Indian Diabetes PIMA dataset. The study emphasized that choosing the appropriate distance measure greatly affects classification accuracy, and

selecting the optimal K value helps eliminate problems of overfitting and underfitting. However, the algorithm’s computational complexity increases with dataset size, as it requires calculating distances to all training instances for each prediction.

2.3.3 XGBoost

XGBoost (eXtreme Gradient Boosting) has emerged as one of the most powerful algorithms for diabetes prediction, consistently achieving state-of-the-art results. A 2024 study using XGBoost with Recursive Feature Elimination (RFE) achieved an impressive 96.4% accuracy, 95% precision, 96% recall, 95.5% F1-score, and 0.98 AUC, demonstrating a +17% accuracy and +15% F1-score enhancement over norm models such as Logistic Regression and Support Vector Machine. The study emphasized that XGBoost’s capability to identify highly relevant predictors, including HbA1c level and blood glucose, combined with its scalability and built-in regularization, makes it highly suitable for clinical field usage.

A 2025 study employing Bayesian optimization to fine-tune XGBoost hyperparameters achieved 97.26% accuracy, 95.72% F1-score, and 81.18% MCC (Matthews Correlation Coefficient). The study demonstrated that Bayesian optimization efficiently locates global optima through a model-based approach, making it more efficient than traditional grid search or random search methods.

Multiple studies have consistently ranked XGBoost among top-performing algorithms. A 2025 comparison study found that XGBoost achieved the highest overall predictive performance with AUC of 0.8168, followed by Random Forest and Logistic Regression (AUCs around 0.79). The model demonstrated approximately 85% accuracy while maintaining high specificity (97%). Another 2025 study leveraging XGBoost and explainable AI techniques achieved 96.07% accuracy and 99.29% AUC, outperforming CatBoost and demonstrating substantial improvement with SMOTE balancing.

XGBoost’s effectiveness stems from several algorithmic advantages: parallel processing capabilities that accelerate training, built-in handling of missing values, L1 and L2 regularization to prevent overfitting, and tree pruning that optimizes model complexity. Feature importance analysis reveals that HbA1c, fasting blood sugar, glucose levels, and BMI consistently rank as top predictors in XGBoost models.

2.3.4 Bayesian Optimized TabNet

Bayesian Optimized TabNet (BO-TabNet) represents a recent advancement in deep learning for tabular healthcare data, specifically designed to provide both high accuracy and interpretability. A seminal 2022 study developed a hybrid Bayesian optimized explainable TabNet model that achieved remarkable accuracy rates of 92.2% on the Pima Indians Diabetes Dataset (PIDD) and 99.4% on the Early-Stage Diabetes Risk Prediction Dataset (ESDRPD). The model significantly surpassed benchmark classifiers including fully convolutional networks (FCN), deep neural networks (DNN), and traditional machine learning models.

TabNet’s architecture incorporates a sequential attention mechanism that enables both local and global interpretability specific to the model. This attention mechanism allows the model to selectively focus on the most relevant features for each prediction, mimicking clinical decision-making processes. XAI analyses revealed that Insulin (importance value: 0.301) was the most significant attribute for diabetes classification from PIDD, while Polyuria (importance value: 0.206) was most significant for ESDRPD. The study employed LIME and SHAP as robust explainable AI tools to enhance model interpretability further.

Bayesian optimization was utilized to efficiently tune TabNet’s numerous hyperparameters, addressing the challenge that traditional methods like grid search pose due to being time-consuming. A 2023 study on blood glucose level prediction using TabNet reported that the model achieved the best performance in terms of RMSE (0.5097) and MSE (0.2523) compared to Logistic Regression, Decision Tree, SVM, Random Forest, and Elastic Net algorithms.

2.3.5 Ensemble Methods

Ensemble learning combines multiple models to improve predictive performance. Voting classifiers that aggregate predictions from multiple base learners have demonstrated improved robustness. A 2025 study reported that a weighted voting ensemble achieved an F2-score of 0.72, though this did not significantly exceed the performance of the best individual model (XGBoost with F2-score of 0.69).

Stacking ensembles use a meta-learner to combine base model predictions. However, the added complexity often does not translate to substantial performance gains,

particularly when the base models are already well-optimized. The risk of overfitting and increased computational requirements are important considerations when deploying ensemble methods in clinical settings.

2.4 Explainable Artificial Intelligence (XAI) Methods

The deployment of machine learning models in healthcare necessitates explainability to build trust, enable clinical validation, and meet regulatory requirements. Explainable AI (XAI) methods provide insights into model predictions, helping clinicians understand which features drive risk assessments.

2.4.1 SHAP (SHapley Additive exPlanations)

SHAP has emerged as one of the most widely adopted XAI methods due to its solid theoretical foundation in cooperative game theory. SHAP values quantify each feature's contribution to a prediction by computing the average marginal contribution across all possible feature combinations. For diabetes prediction models, SHAP analysis consistently identifies blood pressure, waist circumference, triglycerides, and body weight as top predictors, aligning with established clinical knowledge.

A 2024 study analyzing diabetes prediction showed that glucose had the highest mean absolute SHAP value, followed by BMI, indicating these features had the strongest overall impact on predictions. SHAP summary plots (beeswarm plots) show the distribution of SHAP values for each feature across all instances, with color indicating feature values. A 2024 study's SHAP summary plot revealed that high glucose levels (red dots) consistently had positive SHAP values, pushing predictions toward diabetes, while low glucose levels (blue dots) had negative SHAP values, reducing diabetes risk.

SHAP waterfall plots visualize how each feature contributes to moving the prediction from the base value to the final prediction for a specific individual. A 2025 study demonstrated that for a high-risk patient, the waterfall plot showed BMI contributing +0.15 to the risk score, glucose adding +0.22, and age adding +0.08, while physical activity contributed -0.05, collectively moving the prediction to 0.85 (high risk). This allows clinicians to identify which modifiable factors could most effectively reduce a patient's risk.

SHAP dependence plots show the relationship between a feature's value and its

SHAP value, revealing non-linear relationships and interactions. A 2024 study’s SHAP dependence plot for glucose showed that the relationship was approximately linear up to 120 mg/dL, but became more pronounced above this threshold, indicating that higher glucose levels have exponentially greater impact on diabetes risk.

A key advantage of SHAP is its model-agnostic nature, allowing it to explain any machine learning model. TreeExplainer, an optimized implementation for tree-based models, provides exact SHAP values efficiently. However, SHAP can be computationally expensive for large datasets and complex models, and the explanations may be unstable for highly correlated features.

2.4.2 LIME (Local Interpretable Model-agnostic Explanations)

LIME generates local explanations by approximating the model’s behavior in the vicinity of a specific instance using an interpretable surrogate model (typically linear regression). A 2025 study implementing LIME for diabetes prediction showed that for a specific prediction, LIME identified that BMI ≥ 30 contributed +0.25 toward diabetes risk, age ≥ 45 contributed +0.15, while regular exercise contributed -0.10 away from risk.

While LIME is computationally efficient and easy to implement, it suffers from instability—small perturbations in the input can lead to significantly different explanations. This lack of consistency is a critical limitation for clinical applications where reliable explanations are essential for decision-making.

2.4.3 Integration of Calibration with XAI

Several studies have emphasized integrating calibration assessment with XAI techniques. A 2025 study employed SHAP analysis alongside calibration plots to ensure that not only were predictions interpretable, but they were also well-calibrated to reflect true probabilities. The study found that glucose (SHAP value: 2.34) and BMI (SHAP value: 1.87) were primary predictors, demonstrating strong clinical concordance, while calibration plots confirmed that predicted probabilities aligned with observed outcomes.

2.5 Calibrated Explanations

Calibrated Explanations (CE) represents a novel approach to explainable AI that addresses critical limitations of traditional XAI methods by integrating probability cali-

bration directly into the explanation framework.

2.5.1 Foundations of Calibrated Explanations

Calibrated Explanations is built on the foundation of Venn-Abers calibration, a distribution-free method that provides well-calibrated probability estimates. Unlike traditional XAI methods that explain predictions from potentially poorly calibrated models, CE ensures that both the predictions and their explanations are calibrated. This means that when a model predicts a 70% diabetes risk, approximately 70% of similar cases will actually have diabetes.

The method addresses a fundamental problem in existing XAI approaches: feature importance explanations are often unreliable when the underlying model produces mis-calibrated probabilities. By calibrating the model first, CE provides more trustworthy feature importance scores with uncertainty quantification.

2.5.2 Key Features of Calibrated Explanations

CE provides several advantages over traditional XAI methods:

1. **Uncertainty Quantification:** CE provides confidence intervals for feature importance weights, allowing users to assess the reliability of explanations. This is particularly valuable in clinical settings where understanding the certainty of risk factors is crucial.
2. **Factual and Counterfactual Explanations:** CE supports both factual explanations (why did the model make this prediction?) and counterfactual explanations (what changes would alter the prediction?). Counterfactual explanations are especially valuable for patient counseling, as they can identify actionable interventions.
3. **Conditional Rules:** CE generates easily interpretable conditional rules that explain predictions in human-understandable terms, making them accessible to clinicians without machine learning expertise.
4. **Stability and Robustness:** Evaluation on 25 benchmark datasets demonstrated that CE provides more stable explanations compared to LIME and SHAP, with less sensitivity to small perturbations in input features.

2.5.3 Extensions and Applications

Recent extensions of Calibrated Explanations have broadened its applicability:

Calibrated Explanations for Regression extends the framework to support both standard regression and probabilistic regression tasks, where the goal is to predict the probability that a target exceeds a threshold. This is particularly relevant for diabetes prediction when working with continuous biomarkers like HbA1c or fasting blood sugar.

Fast Calibrated Explanations incorporates perturbation techniques to achieve significant computational speedups while maintaining uncertainty quantification. This makes CE suitable for real-time clinical applications where rapid risk assessment is required.

Ensured Explanations addresses epistemic uncertainty by providing guidance on how to reduce prediction uncertainty through targeted feature modifications. This extension categorizes uncertain explanations into counter-potential, semi-potential, and super-potential scenarios, helping clinicians identify the most reliable predictions and understand when additional testing may be needed.

2.5.4 Comparison with Traditional XAI Methods

While SHAP and LIME focus solely on explaining model predictions, Calibrated Explanations integrates calibration and explanation into a unified framework. This distinction is critical: traditional XAI methods can provide misleading explanations when applied to poorly calibrated models, whereas CE ensures that explanations are grounded in well-calibrated probability estimates.

However, it is important to note that Calibrated Explanations is a specific XAI method distinct from the approach of applying calibration techniques (such as Platt Scaling) to a model and then using separate XAI methods (like SHAP or LIME) for explanation. The latter approach, which involves two independent steps, is more common in practice and allows flexibility in choosing calibration and explanation methods separately.

2.6 Probability Calibration Techniques

Probability calibration is essential for ensuring that predicted probabilities accurately reflect true likelihoods, which is critical for clinical decision-making. Miscalibrated mod-

els may produce overconfident or underconfident predictions, leading to inappropriate treatment decisions.

2.6.1 Importance of Calibration in Clinical Settings

Calibration ensures that when a model predicts a patient has a 30% probability of developing diabetes, approximately 30% of such patients actually develop the condition. A 2024 study emphasized that calibration is as important as discrimination for clinical utility, as miscalibrated models can lead to inappropriate treatment decisions even when discrimination is excellent. The study found that a deployed malnutrition prediction model, despite having a c-index of 0.81, was miscalibrated with a calibration intercept of -1.17 and Brier score of 0.26, indicating the model systematically overestimated risks.

Poor calibration has serious consequences in diabetes care. If a prediction model systematically underestimates diabetes risk (calibration intercept < 0), patients who actually need intensive interventions may not receive them, leading to preventable complications. Conversely, overestimation (calibration intercept > 0) leads to unnecessary interventions, anxiety, and healthcare costs.

2.6.2 Calibration Metrics

Multiple metrics assess different aspects of calibration:

Calibration-in-the-large (Calibration Intercept) indicates systematic over- or underestimation across the model, with ideal value of 0; values > 0 suggest overestimation, while values < 0 suggest underestimation. A 2024 study recalibrating a pre-diabetes/diabetes model reported that simple recalibration improved the calibration intercept from -0.56 to -0.01.

Calibration Slope represents the steepness of the calibration curve, with ideal value of 1; values > 1 indicate predictions are too extreme (overconfident), while values < 1 indicate predictions are too moderate. A 2024 validation study found that a diabetes prediction model had a calibration slope of 0.72, suggesting predictions were too extreme.

Brier Score measures overall prediction accuracy by calculating the mean squared difference between predicted probabilities and actual outcomes (0 or 1), with values ranging from 0 (perfect) to 1 (worst). A 2024 study reported Brier scores of 0.26 before recalibration and 0.21 after recalibration, indicating improvement.

Expected Calibration Error (ECE) divides predictions into bins and calculates the weighted average of the absolute difference between confidence and accuracy in each bin. A 2022 study achieving ECE of 0.0018 with LightGBM demonstrated excellent calibration.

2.6.3 Platt Scaling

Platt Scaling, also known as logistic calibration, fits a logistic regression model to map uncalibrated scores to calibrated probabilities. The method is particularly effective for models that produce sigmoid-shaped miscalibration. A 2024 study on diabetes prediction reported a 14.7% improvement in Brier Score after applying Platt Scaling to an XGBoost model, with Log Loss decreasing from 0.6653 to 0.5872. Platt Scaling requires a separate calibration set and is computationally efficient, making it suitable for large-scale applications.

2.6.4 Isotonic Regression

Isotonic Regression is a non-parametric calibration method that fits a piecewise-constant, monotonically increasing function to map uncalibrated scores to calibrated probabilities. Unlike Platt Scaling, Isotonic Regression makes no assumptions about the form of miscalibration, making it more flexible. However, Isotonic Regression is more prone to overfitting, especially with small calibration sets, and may produce less smooth calibration curves.

2.6.5 Venn-Abers Calibration

Venn-Abers is a distribution-free calibration method based on conformal prediction that provides probabilistic predictions with theoretical guarantees. Unlike Platt Scaling and Isotonic Regression, Venn-Abers produces two probability estimates (lower and upper bounds) for each prediction, quantifying the uncertainty in the calibrated probability itself. Venn-Abers is particularly valuable when uncertainty quantification is critical, as it provides confidence intervals for probability estimates.

2.7 Evaluation Metrics for Imbalanced Datasets

Diabetes prediction is inherently an imbalanced classification problem, with diabetic cases typically representing a minority class. Traditional accuracy metrics can be misleading in such scenarios, necessitating the use of specialized evaluation metrics.

2.7.1 Limitations of Accuracy

Accuracy, defined as the proportion of correct predictions among total predictions, is misleading when class distributions are skewed. A 2024 study analyzing diabetes prediction on the BRFSS dataset demonstrated that models could achieve 82-83% accuracy while having severely imbalanced performance metrics: precision (51-75%), recall (7-16%), and F1-score (13-26%). This inconsistency occurred because the model predominantly predicted the majority (non-diabetic) class, achieving high overall accuracy despite failing to identify diabetic cases.

2.7.2 Recall (Sensitivity)

Recall, calculated as $TP/(TP+FN)$, measures the proportion of actual positive cases correctly identified. In diabetes screening, high recall is critical because false negatives—failing to identify individuals with or at risk for diabetes—can lead to missed early intervention opportunities and progression to severe complications. A 2025 study noted that in prediction tasks where minimizing false negatives is critical, models should optimize for recall even if it somewhat reduces precision. The study’s proposed model achieved 88.11% recall, demonstrating strong capability to identify diabetic cases.

2.7.3 Precision

Precision, calculated as $TP/(TP+FP)$, measures the proportion of positive predictions that are correct. High precision minimizes false positives, reducing unnecessary clinical follow-up and patient anxiety. A 2024 study achieving 89.08% precision demonstrated that the model rarely misclassified non-diabetic individuals as diabetic, preventing unnecessary interventions and healthcare costs.

2.7.4 F1-Score and F2-Score

The F1-score, the harmonic mean of Precision and Recall, provides a balanced measure. However, the F2-score, which weights Recall twice as heavily as Precision, is often more appropriate for medical screening applications. A 2024 study on Philippine diabetes data reported an F2-score of 0.69 for an optimized XGBoost model, with Recall of 80.1% and Precision of 44.9%, indicating strong case detection capability suitable for population-level screening.

2.7.5 Area Under the Precision-Recall Curve (AUPRC)

AUPRC is particularly informative for imbalanced datasets, as it focuses on the performance of the positive class across different decision thresholds. Unlike ROC-AUC, which can be misleadingly optimistic for imbalanced data, AUPRC provides a more realistic assessment of model performance. Studies recommend AUPRC over ROC-AUC for diabetes prediction, as it better captures the trade-off between Precision and Recall.

2.7.6 G-means

G-means (geometric mean of sensitivity and specificity) provides a balanced measure particularly suited for imbalanced datasets. A 2018 study using SMOTE with AdaBoost achieved G-mean of 94.65% and AUC of 0.9817 on diabetes follow-up data. G-means is advantageous because it requires both sensitivity and specificity to be high; if either is low, G-means will be low, preventing models from achieving high scores by simply predicting the majority class.

3 Theoretical Framework

3.1 Diabetes and Prediabetes

Diabetes mellitus is a chronic metabolic disorder characterized by elevated blood glucose levels resulting from defects in insulin production, insulin action, or both. Insulin, a hormone produced by pancreatic beta cells, plays a crucial role in regulating blood glucose by facilitating cellular glucose uptake and utilization for energy. When this regulatory mechanism fails, hyperglycemia occurs, leading to both acute and chronic complications affecting multiple organ systems.

The two primary types of diabetes are Type 1 and Type 2 diabetes. Type 1 diabetes is an autoimmune condition in which the immune system destroys insulin-producing beta cells in the pancreas, resulting in absolute insulin deficiency. This form typically manifests in childhood or adolescence and requires lifelong insulin therapy. Type 2 diabetes, which accounts for approximately 90-95% of all diabetes cases, is characterized by insulin resistance and relative insulin deficiency. In this condition, cells fail to respond adequately to insulin, and the pancreas eventually cannot produce sufficient insulin to overcome this resistance.

Prediabetes represents an intermediate metabolic state where blood glucose levels are elevated above normal but below the diagnostic threshold for diabetes. According to the American Diabetes Association, prediabetes is defined by fasting plasma glucose levels of 100-125 mg/dL, or HbA1c levels of 5.7-6.4%. Individuals with prediabetes face significantly increased risk of progressing to Type 2 diabetes, with annual conversion rates of 5-10%. However, prediabetes is reversible through lifestyle interventions, including dietary modifications, increased physical activity, and weight management, making early detection critical for diabetes prevention.

The pathophysiology of Type 2 diabetes involves complex interactions between genetic predisposition and environmental factors. Key risk factors include obesity (particularly central adiposity), physical inactivity, family history of diabetes, age over 45 years, hypertension, dyslipidemia, and certain ethnicities with higher genetic susceptibility. The progression from normal glucose tolerance to prediabetes and eventually to Type 2 diabetes follows a continuum of deteriorating beta-cell function and worsening insulin resistance.

Untreated diabetes leads to serious complications affecting multiple organ systems. Microvascular complications include diabetic retinopathy (leading cause of blindness in working-age adults), diabetic nephropathy (leading cause of end-stage renal disease), and diabetic neuropathy (causing pain, sensory loss, and increased risk of foot ulcers and amputations). Macrovascular complications include accelerated atherosclerosis, increasing risk of coronary artery disease, stroke, and peripheral arterial disease. These complications underscore the critical importance of early detection and intervention to prevent or delay disease progression.

3.2 Machine Learning and Binary Classification

Machine learning is a subfield of artificial intelligence that focuses on developing algorithms capable of learning patterns from data and making predictions or decisions without being explicitly programmed for specific tasks. Machine learning algorithms improve their performance through experience, iteratively refining their models as they process more data. This capability makes machine learning particularly valuable for complex prediction tasks where traditional rule-based approaches are insufficient.

Machine learning paradigms are categorized into three main types based on the learning approach: supervised learning, unsupervised learning, and reinforcement learning. Supervised learning, the focus of this study, involves training models on labeled datasets where both input features and corresponding output labels are provided. The model learns the mapping function from inputs to outputs, enabling it to make predictions on new, unseen data. Unsupervised learning, in contrast, works with unlabeled data to discover hidden patterns or structures, such as clustering similar instances. Reinforcement learning involves agents learning optimal behaviors through trial-and-error interactions with an environment, receiving rewards or penalties based on their actions.

Within supervised learning, tasks are further divided into classification and regression. Classification tasks involve predicting discrete categorical labels, while regression tasks predict continuous numerical values. Diabetes risk prediction is formulated as a binary classification problem, where the goal is to categorize individuals into one of two classes: diabetic (positive class) or non-diabetic (negative class). This binary outcome aligns with clinical decision-making, where individuals are classified based on whether their blood glucose levels exceed diagnostic thresholds.

3.2.1 Traditional Machine Learning Algorithms

Several traditional machine learning algorithms are commonly employed for binary classification tasks in healthcare:

Naive Bayes is a probabilistic classifier based on Bayes' theorem, which calculates the probability of a class given the input features. The "naive" assumption is that all features are conditionally independent given the class label, simplifying computation. Despite this strong assumption, which is often violated in real-world data, Naive Bayes performs surprisingly well in many applications due to its simplicity and computational efficiency. For diabetes prediction, Naive Bayes calculates the probability of diabetes given clinical and demographic features, assigning the class with the highest posterior probability.

Logistic Regression is a linear model for binary classification that estimates the probability of class membership using the logistic (sigmoid) function. Unlike linear regression, which predicts continuous values, logistic regression outputs probabilities constrained between 0 and 1. The model learns a set of weights for each feature, and the weighted sum of features is transformed through the sigmoid function to produce a probability. A decision threshold (typically 0.5) is applied to convert probabilities to binary predictions. Logistic regression is interpretable, as the learned weights indicate the direction and magnitude of each feature's influence on the outcome.

Support Vector Machines (SVM) find an optimal hyperplane that maximally separates instances of different classes in a high-dimensional feature space. The hyperplane is positioned to maximize the margin—the distance between the hyperplane and the nearest data points from each class, called support vectors. SVM can handle non-linear decision boundaries through the kernel trick, which implicitly maps data to higher-dimensional spaces where linear separation becomes possible. Common kernels include linear, polynomial, and radial basis function (RBF). SVM is effective for high-dimensional data but can be computationally expensive for large datasets.

Random Forest is an ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes (for classification) or mean prediction (for regression) of individual trees. Each tree is trained on a bootstrap sample of the data, and at each node, a random subset of features is considered for splitting. This randomness reduces overfitting and improves generalization. Random Forest provides

feature importance scores based on how much each feature decreases impurity across all trees, aiding interpretability. The algorithm handles non-linear relationships, missing values, and mixed data types effectively.

K-Nearest Neighbors (k-NN) is a non-parametric, instance-based learning algorithm that classifies instances based on the majority class among their k nearest neighbors in the feature space. The algorithm stores the entire training dataset and, for each prediction, calculates distances (typically Euclidean) between the query instance and all training instances. The k nearest instances are identified, and the most frequent class among them is assigned as the prediction. The choice of k is critical: small k values lead to complex decision boundaries susceptible to noise, while large k values produce smoother boundaries but may overlook local patterns. k-NN requires feature scaling to prevent features with larger ranges from dominating distance calculations.

Gradient Boosting Machines (XGBoost, LightGBM, CatBoost) are ensemble methods that build models sequentially, where each new model corrects errors made by previous models. XGBoost (Extreme Gradient Boosting) uses gradient descent to minimize a loss function, incorporating regularization to prevent overfitting. LightGBM employs a histogram-based approach and grows trees leaf-wise rather than level-wise, improving efficiency. CatBoost handles categorical features natively and uses ordered boosting to reduce overfitting. These algorithms consistently achieve state-of-the-art performance on tabular data and provide feature importance rankings.

3.2.2 Handling Class Imbalance

Class imbalance occurs when one class significantly outnumbers the other in the training dataset. In diabetes prediction, non-diabetic cases typically far exceed diabetic cases, creating imbalance. Standard machine learning algorithms trained on imbalanced data tend to be biased toward the majority class, achieving high overall accuracy while failing to detect minority class instances—a critical failure in medical screening where missing positive cases (false negatives) has serious consequences.

Several techniques address class imbalance:

Resampling Methods modify the training dataset to balance class distributions. Oversampling increases minority class instances by duplicating existing samples or generating synthetic samples. The Synthetic Minority Over-sampling Technique (SMOTE)

creates synthetic instances by interpolating between minority class instances and their nearest neighbors. Undersampling reduces majority class instances, but risks losing valuable information. Hybrid methods like SMOTE-ENN combine oversampling with edited nearest neighbors to remove noisy instances.

Cost-Sensitive Learning assigns different misclassification costs to different classes, penalizing false negatives more heavily than false positives. Many algorithms support class weights, allowing users to specify the relative importance of each class during training.

Threshold Adjustment modifies the decision threshold for converting predicted probabilities to class labels. Lowering the threshold increases sensitivity (recall) at the cost of precision, which is appropriate for screening applications where detecting all positive cases is prioritized.

3.3 Neural Networks and Optimization Strategies

Neural networks are computational models inspired by the structure and function of biological neural networks in the human brain. They consist of interconnected layers of artificial neurons (nodes) that process information through weighted connections. Each neuron receives inputs, applies a weighted sum, adds a bias term, and passes the result through an activation function to produce an output. Neural networks learn by adjusting weights and biases to minimize a loss function that quantifies prediction errors.

A typical neural network architecture consists of an input layer (receiving features), one or more hidden layers (performing transformations), and an output layer (producing predictions). Deep neural networks contain multiple hidden layers, enabling them to learn hierarchical representations of increasing complexity. Activation functions introduce non-linearity, allowing networks to model complex relationships. Common activation functions include ReLU (Rectified Linear Unit), sigmoid, and tanh.

3.3.1 TabNet Architecture

TabNet is a deep learning architecture specifically designed for tabular data, addressing limitations of traditional neural networks on structured datasets. Unlike standard fully-connected networks that process all features equally, TabNet employs a sequential attention mechanism to select relevant features at each decision step, mimicking human

decision-making processes.

The TabNet encoder consists of several key components:

Feature Transformer processes selected features through fully-connected layers with batch normalization and GLU (Gated Linear Unit) activations, extracting meaningful representations.

Attentive Transformer generates feature selection masks using a sparsemax activation function, which produces sparse outputs where most features have zero weight. This sparsity enhances interpretability by clearly indicating which features influenced each decision step.

Feature Masking applies the selection masks to the input features, allowing only selected features to influence subsequent processing. Masks are updated at each step based on previously selected features, preventing redundant feature selection.

Split Block divides the processed representation into two parts: one for the attentive transformer in the next step, and one contributing to the final output.

TabNet’s sequential attention mechanism provides both instance-wise and global interpretability. Instance-wise interpretability is achieved through visualization of feature selection masks for individual predictions, showing which features were most important. Global interpretability is obtained by aggregating masks across all instances to derive feature importance rankings. This interpretability is crucial for clinical applications where understanding model decisions builds trust and enables validation against medical knowledge.

3.3.2 Bayesian Optimization for Hyperparameter Tuning

Hyperparameters are configuration settings that control the learning process but are not learned from data (e.g., learning rate, number of layers, regularization strength). Selecting optimal hyperparameters significantly impacts model performance, but the search space is often large and high-dimensional, making exhaustive search infeasible.

Bayesian Optimization is an efficient strategy for hyperparameter tuning that models the objective function (e.g., validation performance) as a probabilistic surrogate model, typically a Gaussian Process. The surrogate model provides a posterior distribution over the objective function, quantifying uncertainty about performance at untested hyperparameter configurations.

An acquisition function guides the search by balancing exploration (testing uncertain regions) and exploitation (testing regions likely to yield good performance). Common acquisition functions include Expected Improvement (EI), which selects configurations expected to improve over the current best, and Upper Confidence Bound (UCB), which trades off mean performance and uncertainty.

Bayesian Optimization iteratively:

1. Evaluates the objective function at a selected hyperparameter configuration
2. Updates the surrogate model with the new observation
3. Uses the acquisition function to select the next configuration to evaluate

This approach is particularly effective when objective function evaluations are expensive (as in training deep neural networks), as it requires fewer evaluations than grid search or random search to find near-optimal configurations.

3.4 Explainable Artificial Intelligence and Model Interpretability

Explainable Artificial Intelligence (XAI) refers to methods and techniques that make machine learning model predictions understandable to humans. As models become more complex—particularly deep neural networks—they often function as “black boxes,” producing accurate predictions without transparent reasoning. This opacity is problematic in high-stakes domains like healthcare, where clinicians need to understand and validate model decisions before acting on them.

Model interpretability serves several critical functions: (1) building trust by revealing how models make decisions, (2) enabling clinical validation by allowing experts to verify that models use medically sound reasoning, (3) identifying biases or errors in model logic, (4) facilitating regulatory compliance in domains requiring explainable decisions, and (5) providing actionable insights for interventions by identifying modifiable risk factors.

XAI techniques are categorized along several dimensions:

Scope: Global explanations describe overall model behavior across all instances, while local explanations focus on individual predictions.

Model-specificity: Model-specific techniques are designed for particular algorithms (e.g., decision tree visualization), while model-agnostic techniques can explain any model.

Complexity: Intrinsically interpretable models (e.g., linear regression, decision trees) are transparent by design, while post-hoc explanations are applied to complex models after training.

3.4.1 SHAP (SHapley Additive exPlanations)

SHAP is a unified framework for interpreting model predictions based on Shapley values from cooperative game theory. Shapley values fairly distribute a total payout among players based on their marginal contributions to all possible coalitions. In the context of machine learning, features are "players," and the "payout" is the prediction for a specific instance.

The SHAP value for a feature quantifies its contribution to the prediction by computing the average marginal contribution of that feature across all possible subsets of features. Mathematically, for feature i and instance x :

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)]$$

where F is the set of all features, S is a subset of features, and $f(S)$ is the model's prediction using only features in S .

SHAP values have desirable properties: (1) local accuracy—SHAP values sum to the difference between the prediction and the expected value, (2) missingness—features not used in the model have zero SHAP value, (3) consistency—if a model changes so a feature has a larger impact, its SHAP value increases.

SHAP provides multiple visualization types:

Force plots show how each feature pushes the prediction away from the base value (average prediction) toward the final prediction for a single instance.

Summary plots display SHAP values for all features across all instances, revealing global feature importance and the distribution of impacts.

Dependence plots show the relationship between feature values and SHAP values, revealing non-linear effects and interactions.

TreeExplainer is an optimized SHAP implementation for tree-based models that computes exact SHAP values efficiently. For other models, KernelExplainer approximates SHAP values through sampling.

3.4.2 LIME (Local Interpretable Model-agnostic Explanations)

LIME explains individual predictions by approximating the complex model’s behavior locally with a simple, interpretable surrogate model. The key insight is that while a model may be globally complex, its behavior in the vicinity of a specific instance may be approximated by a linear model.

LIME’s algorithm:

1. Generates a dataset of perturbed instances around the instance to be explained
2. Obtains predictions from the complex model for these perturbed instances
3. Weights instances by their proximity to the original instance
4. Trains a simple interpretable model (e.g., linear regression) on the weighted dataset
5. Uses the interpretable model’s coefficients as the explanation

For tabular data, LIME perturbs instances by randomly sampling from the training distribution for each feature. For text and images, perturbations involve removing words or image segments.

LIME’s explanations are local—they describe model behavior for a specific instance and may not generalize to other instances. This locality is both a strength (providing personalized explanations) and a limitation (not revealing global model behavior). LIME is also subject to instability, where small changes in the instance or perturbation process can yield different explanations.

3.5 Probability Calibration Methods

Probability calibration is the process of adjusting predicted probabilities from a machine learning model to better reflect true likelihoods of outcomes. Many classification algorithms, particularly complex models like neural networks and ensemble methods, produce miscalibrated probability estimates—predictions that do not accurately represent the true probability of class membership. For instance, when a model predicts a 70% probability of diabetes, ideally 70% of such cases should actually be diabetic. Miscalibration can lead to inappropriate clinical decisions, as healthcare providers rely on these probabilities to determine intervention strategies.

Calibration is distinct from discrimination (the ability to rank instances correctly). A model can have excellent discrimination (high AUC-ROC) while being poorly calibrated. Calibration ensures that predicted probabilities are reliable and can be used directly for decision-making, which is critical in medical applications where probability estimates guide treatment decisions.

3.5.1 Platt Scaling

Platt Scaling, also known as logistic calibration, is a parametric calibration method that fits a logistic regression model to map uncalibrated scores to calibrated probabilities. The method was originally developed for Support Vector Machines but is applicable to any classifier that produces continuous scores.

Given uncalibrated scores $f(x)$ from a classifier, Platt Scaling learns parameters A and B such that the calibrated probability is:

$$P(y = 1|x) = \frac{1}{1 + \exp(A \cdot f(x) + B)}$$

The parameters A and B are estimated by maximizing the likelihood on a held-out calibration set (separate from the training set used to train the original classifier). This calibration set should be representative of the target population to ensure the calibrated probabilities generalize well.

Platt Scaling is particularly effective when the miscalibration follows a sigmoid pattern—when uncalibrated scores have a monotonic but non-linear relationship with true probabilities. The method is computationally efficient, requiring only the fitting of two parameters, and works well even with relatively small calibration sets. However, Platt Scaling assumes a specific functional form (sigmoid), which may not capture all types of miscalibration patterns.

3.5.2 Isotonic Regression

Isotonic Regression is a non-parametric calibration method that fits a piecewise-constant, monotonically increasing function to map uncalibrated scores to calibrated probabilities. Unlike Platt Scaling, Isotonic Regression makes no assumptions about the functional form of the calibration mapping, making it more flexible.

The method works by:

1. Sorting instances by their uncalibrated scores
2. Grouping instances into bins
3. Computing the empirical probability (fraction of positive instances) in each bin
4. Enforcing monotonicity by merging adjacent bins that violate the monotonicity constraint

The resulting calibration function is a step function that maps score ranges to calibrated probabilities. For a new instance with score $f(x)$, the calibrated probability is the empirical probability of the bin containing $f(x)$.

Isotonic Regression is more flexible than Platt Scaling and can capture complex miscalibration patterns, including non-sigmoid relationships. However, this flexibility comes at a cost: Isotonic Regression is more prone to overfitting, especially with small calibration sets, and may produce less smooth calibration curves. The method also requires more data than Platt Scaling to reliably estimate the piecewise-constant function.

3.5.3 Calibration Assessment

Calibration quality is assessed through several metrics and visualizations:

Calibration Plots (Reliability Diagrams) plot predicted probabilities against observed frequencies. Instances are grouped into bins based on predicted probability, and for each bin, the mean predicted probability is plotted against the fraction of positive instances. A perfectly calibrated model produces points along the diagonal line $y = x$.

Brier Score measures the mean squared difference between predicted probabilities and actual outcomes:

$$\text{Brier Score} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$$

where p_i is the predicted probability and y_i is the actual outcome (0 or 1). Lower Brier scores indicate better calibration and discrimination. The Brier score ranges from 0 (perfect) to 1 (worst).

Expected Calibration Error (ECE) divides predictions into bins and computes the weighted average of the absolute difference between confidence and accuracy in each

bin:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$$

where B_m is the set of instances in bin m , $\text{acc}(B_m)$ is the accuracy in that bin, and $\text{conf}(B_m)$ is the average predicted probability in that bin. Lower ECE indicates better calibration.

In this study, both Platt Scaling and Isotonic Regression are applied to calibrate the predictions from machine learning models, with Platt Scaling typically preferred due to its robustness and lower risk of overfitting on the available calibration data.

4 Design and Implementation

This chapter presents the design and implementation of the diabetes risk prediction system with calibrated explanations. The methodology encompasses data acquisition and preprocessing, machine learning model development with multiple algorithms, probability calibration techniques, explainable AI integration, and performance evaluation. The chapter is organized into seven major sections: conceptual framework, dataset description, preprocessing techniques, machine learning models, probability calibration methods, explainable AI techniques, and performance evaluation metrics.

4.1 Conceptual Framework

The conceptual framework of this study consists of four key stages: **Input**, **Preprocessing**, **Processing**, and **Output**.

4.1.1 Input Stage

The input stage involves acquiring data from the DOST-FNRI Expanded National Nutrition Survey (ENNS) 2013. Four primary datasets are loaded:

- **Clinical and Health Component Dataset:** Contains clinical measurements including fasting blood sugar (FBS), blood pressure, cholesterol levels, and lifestyle factors. This dataset includes information on smoking status, alcohol consumption patterns, and physical activity levels.
- **Biochemical Component Dataset:** Includes laboratory test results such as lipid profiles (HDL-cholesterol, LDL-cholesterol, triglycerides), providing detailed biochemical indicators of metabolic health.
- **Anthropometric Component Dataset:** Contains body measurements including height, weight, body mass index (BMI), waist circumference, and hip circumference. These measurements are critical for assessing obesity and central adiposity.
- **Dietary Component Dataset:** Provides information on food group consumption (cereals, vegetables, fruits, meat, dairy, etc.) and nutrient intake (energy, protein, carbohydrates, fats, vitamins, minerals).

These datasets are merged using household number (`hhnum`) and member code (`member_code`) as common identifiers to create a comprehensive dataset for diabetes prediction. The initial merged dataset contains 90,286 individuals with 77 features before preprocessing.

4.1.2 Preprocessing Stage

The preprocessing stage applies multiple data transformation techniques to prepare the merged dataset for machine learning:

1. **Data Filtering:** Exclude children (age < 20), pregnant women, and lactating women to focus on the adult non-pregnant population.
2. **Missing Value Handling:** Drop rows with missing target variable (FBS); impute missing numerical features with K-Nearest Neighbors (KNN) imputation; impute missing categorical features with mode or placeholder value.
3. **Target Variable Conversion:** Transform continuous FBS values into binary classification (0 = non-diabetic for FBS < 100 mg/dL; 1 = diabetic for FBS ≥ 100 mg/dL).
4. **Feature Scaling:** Apply MinMaxScaler to all numerical features, transforming them to a range of $[0, 1]$.
5. **Feature Selection:** Remove highly correlated features (correlation > 0.8) to reduce multicollinearity and redundancy.
6. **Class Imbalance Handling:** Apply SMOTE (Synthetic Minority Over-sampling Technique) to the training set to balance diabetic and non-diabetic classes.
7. **Data Leakage Prevention:** Perform train-validation-test split (70-15-15) before applying SMOTE and calibration to prevent data leakage.

4.1.3 Processing Stage

The processing stage involves training multiple machine learning models, applying calibration, and generating explanations:

Model Training: Nine machine learning algorithms are trained on the preprocessed training set:

- Naive Bayes
- K-Nearest Neighbors (k-NN)
- Random Forest
- XGBoost (Extreme Gradient Boosting)
- LightGBM (Light Gradient Boosting Machine)
- CatBoost (Categorical Boosting)
- AdaBoost (Adaptive Boosting)
- TabNet (Bayesian Optimized using validation set)
- Ensemble Methods (Voting Classifier and Stacking Classifier)

Hyperparameter Optimization: For TabNet, Bayesian Optimization is performed using the validation set to find optimal hyperparameters (learning rate, number of decision steps, attention dimensions, etc.).

Probability Calibration: Two calibration methods are applied to each trained model using the validation set:

- Platt Scaling (Logistic Calibration)
- Isotonic Regression

Calibration ensures that predicted probabilities accurately reflect true diabetes risk, enabling reliable risk stratification.

Explainable AI Integration: Two XAI techniques are applied to generate explanations:

- **SHAP (SHapley Additive exPlanations):** Provides global feature importance and individual prediction explanations based on Shapley values from game theory
- **Calibrated Explanations:** Generates feature importance with uncertainty quantification, providing probabilistic explanations that account for model uncertainty

4.1.4 Output Stage

The output stage presents the results of the diabetes risk prediction system:

- **Prediction:** Binary classification (diabetic/non-diabetic) with calibrated probability estimate
- **Risk Stratification:** Classification into risk levels (Low: $\leq 30\%$, Moderate: $30\text{--}70\%$, High: $\geq 70\%$) based on calibrated probabilities
- **Feature Importance:** Ranking of features by their contribution to the prediction
- **Calibrated Explanations:** Feature-level explanations with uncertainty intervals showing which features increase or decrease diabetes risk
- **Performance Metrics:** Model evaluation using Precision, Recall, F2-Score, AUPRC, and calibration metrics (Brier Score, ECE)

4.2 DOST-FNRI 2013 National Nutrition Survey Dataset

The dataset used in this study is sourced from the eNutrition website of the Food and Nutrition Research Institute (FNRI), an attached agency under the Department of Science and Technology (DOST) in the Philippines. The eNutrition website serves as the National Nutrition Survey (NNS) data warehouse, providing electronically accessible data on population health, nutrition, and dietary patterns.

4.2.1 Clinical and Health Component Dataset

This dataset contains 105,949 individuals with 29 features, including:

Demographic Features: Region code, province code, household number, member code, age, sex, physiological status, civil status

Clinical Measurements: Average systolic blood pressure (SBP), average diastolic blood pressure (DBP), fasting blood sugar (FBS - target variable), total cholesterol, triglycerides, HDL-cholesterol, LDL-cholesterol

Lifestyle Factors: Current smoking status, alcohol consumption patterns, physical activity level (MET classification)

Survey Metadata: Primary sampling unit, strata, sampling weights

The FBS feature serves as the target variable, representing fasting blood glucose levels measured after 10-12 hours of fasting. This dataset was designed to assess risk factors for non-communicable diseases (NCDs) including diabetes, hypertension, and dyslipidemia.

4.2.2 Biochemical Component Dataset

This dataset provides laboratory test results for lipid profiles and other biomarkers, complementing the clinical measurements with detailed biochemical indicators of metabolic health.

4.2.3 Anthropometric Component Dataset

This dataset contains body measurements including height, weight, body mass index (BMI), waist circumference, and hip circumference. These anthropometric indicators are critical predictors of diabetes risk, particularly central obesity measured by waist circumference.

4.2.4 Dietary Component Dataset

This dataset contains 19,831 individuals with 48 features representing food group consumption and nutrient intake:

Food Groups (grams/day): Cereals and cereal products, rice, corn, starchy roots and tubers, sugar and syrups, dried beans/nuts/seeds, vegetables (green leafy, yellow, others), fruits (vitamin C-rich, others), fish/meat/poultry, eggs, milk and milk products, fats and oils, beverages, condiments and spices

Nutrient Intake: Total energy (kcal), protein (g), calcium (mg), iron (mg), vitamin A (μ g), thiamin (mg), riboflavin (mg), niacin (mg), vitamin C (mg), carbohydrates (g), fats (g)

This dataset provides insights into dietary patterns and their relationship with diabetes risk, as diet is a major modifiable risk factor for Type 2 diabetes.

4.3 Preprocessing Techniques

All preprocessing was implemented using Python 3.9 with the following libraries: Pandas (version 1.5.3) for data manipulation, NumPy (version 1.24.2) for numerical operations,

and Scikit-learn (version 1.2.2) for preprocessing utilities and machine learning algorithms.

4.3.1 Dataset Merging

The four component datasets (clinical, biochemical, anthropometric, dietary) are merged using a left join on the common identifiers `hhnum` (household number) and `member_code` (member identifier). The clinical dataset serves as the base, as it contains the target variable (FBS). The merge operation combines all available information for each individual into a single comprehensive dataset.

The merged dataset initially contains 90,286 entries with 77 features. These features include demographic information (region code, province code, age, sex, civil status), clinical measurements (blood pressure, FBS, cholesterol, triglycerides, HDL, LDL), anthropometric measurements (height, weight, BMI, waist, hip), lifestyle factors (smoking status, alcohol consumption, physical activity), and dietary intake (food groups and nutrient consumption).

4.3.2 Data Filtering

To ensure the study focuses on the appropriate population, the following exclusion criteria are applied:

- **Age Filter:** Individuals under 20 years old are excluded, as diabetes screening guidelines typically target adults.
- **Pregnancy Status:** Pregnant women are excluded to avoid confounding from gestational diabetes.
- **Lactation Status:** Lactating women are excluded as their metabolic profiles may differ from the general population.

Additionally, administrative and survey design variables are removed to prevent data leakage. Specifically, `regcode` (region code), `provcode` (province code), `psurec` (primary sampling unit), `strrec` (strata), and sampling weight variables (`finalwgt1`, `finalwgt4`) are dropped. These variables are design artifacts that do not represent individual health characteristics and could allow models to learn geographic patterns rather than clinical risk factors.

4.3.3 Missing Value Handling

Missing data is handled using a multi-step approach:

1. **Target Variable:** Rows with missing FBS values are dropped. For the sake of transparency and to preserve the integrity of the data, as well as to avoid imputation bias, the missing target values are dropped instead of imputed. This removes instances where diabetes status cannot be determined.
2. **Features with Excessive Missing Values:** Features with more than 50% missing values are dropped entirely, as imputation would introduce too much uncertainty. For example, `drnk_30dnum` (number of drinks in past 30 days) was removed due to excessive missingness.
3. **Numerical Features:** Missing values are imputed using K-Nearest Neighbors (KNN) imputation with $k=5$ neighbors. This method preserves the local structure of the data better than simple mean imputation. The KNN imputer is fitted on the training set only to prevent data leakage.
4. **Categorical Features:** Missing values are imputed with the mode (most frequent value) calculated from the training set. For some categorical features, missing values are assigned a placeholder value of 99, as indicated in the ENNS metadata, which represents "Not Applicable" or "Unknown."

4.3.4 Target Variable Conversion

The target attribute `fbs` represents the fasting blood sugar level and is inherently a continuous range of values. Since this study is performing binary classification, the target attribute is transformed into a binary categorical variable. The continuous FBS variable is converted to a binary target based on the American Diabetes Association diagnostic criteria:

- **Non-diabetic (Class 0):** $\text{FBS} < 100 \text{ mg/dL}$ (normal fasting glucose)
- **Diabetic (Class 1):** $\text{FBS} \geq 100 \text{ mg/dL}$ (includes both prediabetes [100-125 mg/dL] and diabetes [$\geq 126 \text{ mg/dL}$])

This threshold-based conversion creates a binary classification problem suitable for the machine learning models employed. The combined prediabetes and diabetes class reflects the clinical importance of early detection, as individuals with prediabetes are at high risk of progression to Type 2 diabetes and benefit from early intervention.

4.3.5 Feature Scaling

To scale the features of the merged dataset, the columns are segregated between numerical and categorical values. During feature scaling, only numerical features are scaled, while categorical features are left unchanged.

MinMaxScaler is applied to all numerical features, transforming them to a range of $[0, 1]$:

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

where x is the original feature value, $x_{\min}$ is the minimum value, and $x_{\max}$ is the maximum value in the training set. MinMaxScaler ensures that features with different units and scales (e.g., age in years vs. cholesterol in mg/dL) are normalized to a common range, which is particularly important for distance-based algorithms like k-NN and neural networks like TabNet. The scaling parameters ($x_{\min}$, $x_{\max}$) are computed from the training set only and applied to validation and test sets to prevent data leakage.

4.3.6 Feature Selection

After merging the four datasets, the merged dataset contains 77 features. Feature selection is applied to choose a subset of features while excluding others based on their correlation and relevance. Correlation-based feature selection is performed to remove redundant features:

1. Compute the Pearson correlation matrix for all numerical features
2. Identify pairs of features with correlation coefficient > 0.8
3. For each highly correlated pair, remove the feature with lower correlation to the target variable

Highly correlated features indicate that they carry similar information, which leads to multicollinearity and redundancy. Once highly correlated features are identified, they are removed from the merged dataset. This reduces multicollinearity, improves model interpretability, and decreases computational cost.

After feature selection, the final dataset contains 64 features. These features represent a diverse set of health indicators including demographic factors, clinical measurements, anthropometric indices, lifestyle behaviors, and dietary patterns.

4.3.7 Handling Class Imbalance

The dataset exhibits class imbalance, with non-diabetic cases significantly outnumbering diabetic cases. To address this imbalance, SMOTE (Synthetic Minority Over-sampling Technique) is applied to the training set:

1. For each minority class instance, identify $k=5$ nearest neighbors from the same class
2. Generate synthetic instances by interpolating between the instance and its neighbors using the formula: $x_{\text{synthetic}} = x_i + \lambda \cdot (x_{\text{neighbor}} - x_i)$ where $\lambda \in [0, 1]$ is a random value
3. Continue until the desired class balance is achieved (65:35 majority-to-minority ratio)

Critically, SMOTE is applied only to the training set **after** the train-validation-test split to prevent data leakage. Applying SMOTE before splitting would allow synthetic instances based on test data to appear in the training set, artificially inflating performance metrics. The validation and test sets retain their original class distributions to provide realistic performance estimates that reflect real-world deployment scenarios.

After performing all preprocessing techniques, the dataset is reduced to 90,286 entries with 64 features and 1 target variable. The data is partitioned using a 70-15-15 train-validation-test split. Consequently, the training set contains 67,206 entries, the validation set contains 11,540 entries, and the test set contains 11,540 entries.

4.4 Machine Learning Models

Nine machine learning algorithms are trained and evaluated for diabetes prediction. All models are implemented using Python 3.9 with Scikit-learn (version 1.2.2) for traditional machine learning algorithms and PyTorch-TabNet (version 4.0) for the deep learning model. Additional libraries used include XGBoost (version 1.7.5), LightGBM (version 3.3.5), and CatBoost (version 1.2) for gradient boosting implementations.

4.4.1 Naive Bayes

A Gaussian Naive Bayes classifier is trained using `sklearn.naive_bayes.GaussianNB`, assuming that features follow a normal distribution within each class. Despite the strong independence assumption that all features are conditionally independent given the class label, Naive Bayes provides a fast baseline model with minimal hyperparameter tuning required. The model is trained with default parameters.

4.4.2 K-Nearest Neighbors

A k-Nearest Neighbors classifier is trained using `sklearn.neighbors.KNeighborsClassifier`. Hyperparameter tuning is performed using `RandomizedSearchCV` with 5-fold cross-validation to optimize the number of neighbors (k) and distance metric. The final model uses k=5 neighbors and Euclidean distance metric. The algorithm classifies instances based on the majority vote of their nearest neighbors in the feature space. Due to the instance-based nature of k-NN, feature scaling is critical for this model.

4.4.3 Random Forest

A Random Forest classifier is trained using `sklearn.ensemble.RandomForestClassifier`. Hyperparameter tuning is performed using `RandomizedSearchCV` with 5-fold cross-validation, optimizing for F2-score to prioritize recall. The search space includes: number of estimators (50-500), maximum depth (10-100), minimum samples split (2-20), minimum samples leaf (1-10), and maximum features (sqrt, log2, None). The final tuned model uses 100 decision trees with a maximum depth of 30. Each tree is built on a bootstrap sample of the training data, and random feature subsets are considered at each split. The final prediction is the majority vote across all trees.

4.4.4 Gradient Boosting Models

Three gradient boosting implementations are trained:

XGBoost: Extreme Gradient Boosting is trained using `xgboost.XGBClassifier`. Hyperparameter tuning is performed using `RandomizedSearchCV` with 5-fold cross-validation and 50 iterations, optimizing for F2-score. The search space includes: learning rate (0.01-0.3), maximum depth (3-10), number of estimators (100-1000), subsample ratio (0.6-1.0), and regularization parameters (alpha, lambda). The final tuned model achieves the best performance among all models tested, with tree-based learners, L1 and L2 regularization, and optimized parallel processing.

LightGBM: Light Gradient Boosting Machine is trained using `lightgbm.LGBMClassifier`. The model uses histogram-based algorithms and leaf-wise tree growth for improved efficiency compared to level-wise growth in traditional gradient boosting. Default parameters are used with minor adjustments for class imbalance (`scale_pos_weight`).

CatBoost: Categorical Boosting is trained using `catboost.CatBoostClassifier`. The model provides native categorical feature handling and uses ordered boosting to reduce overfitting. Training is performed with 1000 iterations and early stopping based on validation set performance.

4.4.5 AdaBoost

Adaptive Boosting is trained using `sklearn.ensemble.AdaBoostClassifier`. Hyperparameter tuning is performed using `RandomizedSearchCV` with 5-fold cross-validation. The search space includes: number of estimators (50-500), learning rate (0.01-2.0), and base estimator type. The model trains a sequence of weak learners (decision stumps), with each subsequent learner focusing on instances misclassified by previous learners. Instance weights are adjusted iteratively to emphasize difficult cases, making the ensemble progressively better at handling challenging instances.

4.4.6 TabNet

TabNet is a deep learning architecture for tabular data with sequential attention mechanism, implemented using `pytorch.tabnet.tab_model.TabNetClassifier`. Two versions are trained:

TabNet (Baseline): Trained with default hyperparameters as a baseline for comparison.

TabNet (Bayesian Optimized): Bayesian Optimization is performed using `skopt.BayesSearchC` with 12 iterations and 5-fold cross-validation (60 total model evaluations) to tune hyperparameters. The search space includes:

- `n_d` and `n_a` (decision and attention dimensions): [8, 16, 32, 64]
- `n_steps` (number of decision steps): [3, 4, 5, 6, 7]
- `gamma` (relaxation parameter): [1.0, 1.5, 2.0]
- `lambda_sparse` (sparsity regularization): [1e-6, 1e-4, 1e-3]
- Learning rate: [0.005, 0.01, 0.02]
- Batch size: [256, 512, 1024]

The Bayesian Optimization process took approximately 214 minutes to complete. The optimized TabNet model provides both predictive performance and interpretability through feature selection masks, which show which features were selected at each decision step for individual predictions.

4.4.7 Ensemble Methods

Two ensemble approaches are implemented using `sklearn.ensemble`:

Voting Classifier: Implemented using `VotingClassifier` with soft voting. Combines predictions from multiple base models (Random Forest, XGBoost, LightGBM, CatBoost) by averaging their predicted probabilities. Each base model contributes equally to the final prediction (uniform weights).

Stacking Classifier: Implemented using `StackingClassifier`. Uses predictions from base models (Random Forest, XGBoost, LightGBM, CatBoost, AdaBoost) as features for a meta-learner. The meta-learner is a Logistic Regression model trained using 5-fold cross-validation on the training set. This enables the meta-learner to learn optimal combination weights based on each base model’s strengths and weaknesses.

4.5 Probability Calibration

After training, each model’s predicted probabilities are calibrated using the validation set to ensure they accurately reflect true diabetes risk.

4.5.1 Platt Scaling

Platt Scaling fits a logistic regression model to map uncalibrated scores to calibrated probabilities. For each model, the validation set predictions are used to learn parameters A and B such that:

$$P_{\text{calibrated}}(y = 1|x) = \frac{1}{1 + \exp(A \cdot f(x) + B)}$$

where $f(x)$ is the uncalibrated score from the original model.

4.5.2 Isotonic Regression

Isotonic Regression fits a piecewise-constant monotonic function to map uncalibrated scores to calibrated probabilities. The validation set is used to learn the calibration mapping, which is then applied to test set predictions.

4.5.3 Calibration Evaluation

Calibration quality is assessed using:

- **Calibration Plots:** Visual comparison of predicted probabilities vs. observed frequencies
- **Brier Score:** Mean squared error between predicted probabilities and actual outcomes
- **Expected Calibration Error (ECE):** Average absolute difference between confidence and accuracy across probability bins

4.6 Explainable Artificial Intelligence Techniques

Two XAI techniques are implemented to provide interpretable explanations for model predictions.

4.6.1 SHAP (SHapley Additive exPlanations)

SHAP is implemented using the SHAP Python library. For tree-based models (Random Forest, XGBoost, LightGBM, CatBoost), TreeExplainer computes exact Shapley values efficiently. For other models, KernelExplainer approximates Shapley values through sampling.

SHAP provides:

- **Global Feature Importance:** Ranking of features by mean absolute SHAP value across all instances
- **Local Explanations:** Feature contributions for individual predictions shown through force plots
- **Summary Plots:** Distribution of SHAP values across all instances, revealing feature impact patterns

4.6.2 Calibrated Explanations

Calibrated Explanations are implemented using the `calibrated-explanations` Python library. This technique provides:

- **Probabilistic Feature Importance:** Feature contributions with uncertainty intervals based on conformal prediction
- **Factual Explanations:** Features that support the predicted class
- **Counterfactual Explanations:** Features that, if changed, would alter the prediction
- **Uncertainty Quantification:** Confidence intervals for both predictions and explanations

Calibrated Explanations leverage the calibrated probabilities from Platt Scaling or Isotonic Regression to provide reliable uncertainty estimates.

4.7 Performance Evaluation Metrics

Model performance is evaluated using multiple complementary metrics to assess different aspects of predictive quality.

4.7.1 Confusion Matrix

The confusion matrix summarizes classification performance by tabulating predicted versus actual classes:

- **True Positives (TP):** Diabetic cases correctly predicted as diabetic
- **True Negatives (TN):** Non-diabetic cases correctly predicted as non-diabetic
- **False Positives (FP):** Non-diabetic cases incorrectly predicted as diabetic (Type I error)
- **False Negatives (FN):** Diabetic cases incorrectly predicted as non-diabetic (Type II error)

4.7.2 Precision, Recall, and F-Scores

Precision (Positive Predictive Value) measures the proportion of positive predictions that are correct:

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision indicates few false alarms, minimizing unnecessary follow-up testing and patient anxiety.

Recall (Sensitivity, True Positive Rate) measures the proportion of actual positive cases correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall is critical in medical screening to minimize missed diagnoses (false negatives).

F1-Score is the harmonic mean of precision and recall, providing a balanced measure:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

F2-Score is a weighted harmonic mean that emphasizes recall over precision:

$$F_2 = 5 \cdot \frac{\text{Precision} \cdot \text{Recall}}{4 \cdot \text{Precision} + \text{Recall}}$$

The F2-score is the primary evaluation metric for this study, as it prioritizes recall

(minimizing false negatives) while maintaining reasonable precision. In diabetes screening, the cost of missing a diabetic case (false negative) exceeds the cost of a false alarm (false positive), making F2-score more appropriate than F1-score or accuracy.

4.7.3 Specificity and Accuracy

Specificity (True Negative Rate) measures the proportion of actual negative cases correctly identified:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Accuracy measures overall correctness:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

While accuracy is reported, it is not used as the primary metric due to class imbalance, as a model predicting only the majority class can achieve high accuracy while failing to detect minority class instances.

4.7.4 ROC Curve and AUC

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (Recall) against the False Positive Rate (1 - Specificity) across all classification thresholds. The Area Under the ROC Curve (AUC-ROC) quantifies overall discriminative ability, with values ranging from 0.5 (random guessing) to 1.0 (perfect classification).

However, ROC-AUC can be overly optimistic for imbalanced datasets, as it gives equal weight to both classes.

4.7.5 Precision-Recall Curve and AUPRC

The Precision-Recall (PR) curve plots Precision against Recall across all thresholds. The Area Under the Precision-Recall Curve (AUPRC) is more informative than AUC-ROC for imbalanced datasets, as it focuses on performance on the positive (minority) class. AUPRC is sensitive to improvements in detecting positive cases and serves as a secondary evaluation metric alongside F2-score.

4.7.6 Calibration Metrics

Brier Score measures the mean squared difference between predicted probabilities and actual outcomes:

$$\text{Brier Score} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2$$

where p_i is the predicted probability and y_i is the actual outcome (0 or 1). Lower Brier scores indicate better calibration and discrimination.

Expected Calibration Error (ECE) quantifies calibration quality:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$$

where B_m is the set of instances in bin m , $\text{acc}(B_m)$ is the accuracy in that bin, and $\text{conf}(B_m)$ is the average predicted probability in that bin. Lower ECE indicates better calibration.

5 Expected Output/Timeline

Screenshots of your final software and Gantt chart of your activities are placed here.

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